

LINK COMMANDER / Mouser PO

Five PCB builds — full parts order

PO reference: LC-D-5-FULL-20260918 **Date:** 2026-09-18

Supplier: Mouser Electronics • 1000 North Main Street • Mansfield, TX 76063

Requested fulfillment: WILL-CALL / CUSTOMER PICKUP — do not ship.

Buyer/company: _____ **Account:** _____

Pickup person/phone: _____ **Pickup date:** _____

Draft only. Not submitted, paid or reserved. Mouser must quote prices/tax/total and confirm stock, will-call availability and ready time. Exact parts; contact buyer before substitutions.

Order overlap: The full PO includes these same five of each chip. Submit both only if ten of each chip are intended.

Supply separately: FIVE RP2040-Zero modules and FIVE fabricated Rev D PCBs. Existing workshop tools are assumed. This orders PCB assembly parts, not breadboards.

A. Circuit components

#	Qty	Mouser SKU / Mfr. part	Description / allocation	Unit \$	Ext. \$
1	5	595-TPS2042P TPS2042P	USB power-switch chip — PDIP-8	Quote	Quote
2	5	595-CD4050BE CD4050BE	PS/2 input buffer — PDIP-16	Quote	Quote
3	20	512-2N3904BU 2N3904BU	NPN transistor — TO-92	Quote	Quote
4	20	603-MFR-25FRF52-22R MFR-25FRF52-22R	22 Ω resistor — 1%, ¼ W, axial; select Cut Tape Cut Tape; loose resistors, not a reel	Quote	Quote
5	20	603-MFR-25FBF52-15K MFR-25FBF52-15K	15 k Ω resistor — 1%, ¼ W, axial	Quote	Quote
6	30	603-MFR-25FBF52-2K2 MFR-25FBF52-2K2	2.2 k Ω resistor — 1%, ¼ W, axial	Quote	Quote
7	20	603-MFR-25FBF52-47K MFR-25FBF52-47K	47 k Ω resistor — 1%, ¼ W, axial	Quote	Quote
8	20	603-MFR-25FBF52-4K7 MFR-25FBF52-4K7	4.7 k Ω resistor — 1%, ¼ W, axial	Quote	Quote
9	20	603-MFR-25FBF52-100K MFR-25FBF52-100K	100 k Ω resistor — 1%, ¼ W, axial	Quote	Quote
10	20	594-K104K15X7RF5TH5 K104K15X7RF5TH5	100 nF ceramic capacitor — 50 V, leaded	Quote	Quote
11	5	647-UVR1E471MPD UVR1E471MPD	470 μ F electrolytic — 25 V, radial	Quote	Quote
12	5	667-EEU-FR1H100 EEU-FR1H100	10 μ F electrolytic — 50 V, radial	Quote	Quote
13	10	667-EEU-FR1C221 EEU-FR1C221	220 μ F electrolytic — 16 V, radial	Quote	Quote
14	10	604-WP710A10ID WP710A10ID	Red LED — 3 mm, through-hole	Quote	Quote
15	5	504-BK1/GMA-1.5-R BK1/GMA-1.5-R	1.5 A fast-blow fuse — 5 × 20 mm	Quote	Quote

B. PCB connectors, sockets and hardware

#	Qty	Mouser SKU / Mfr. part	Description / allocation	Unit \$	Ext. \$
16	10	538-67643-3910 67643-3910	USB-A female through-hole socket 2 per PCB; solder directly to board	Quote	Quote
17	10	490-SDS-70J SDS-70J	7-pin DIN female through-hole socket 2 per PCB; solder directly to board	Quote	Quote
18	5	571-282834-2 282834-2	2-position PCB power terminal, 2.54 mm J1; one per PCB	Quote	Quote
19	5	571-1-2199298-2 1-2199298-2	8-pin DIP socket, 7.62 mm row spacing U1 socket; one per PCB	Quote	Quote
20	5	571-1-2199298-4 1-2199298-4	16-pin DIP socket, 7.62 mm row spacing U2 socket; one per PCB	Quote	Quote
21	10	200-SSW10901TS SSW-109-01-T-S	9-contact female module socket, 2.54 mm M1 side rows; two per PCB	Quote	Quote
22	25	200-SSW10101TS SSW-101-01-T-S	Single-contact female module socket Five per PCB at M1 contacts 10-14; see orientation note	Quote	Quote
23	1	485-392 392	Male header strips — ten 36-pin strips, 2.54 mm ONE PACK of ten 36-pin strips; supplies all module male headers and five JP1 headers	Quote	Quote
24	5	571-8815452 881545-2	2-pin removable shunt with handle, 2.54 mm JP1 cap; one per PCB	Quote	Quote
25	10	576-01110501Z 01110501Z	111-series through-hole 5x20 mm fuse clip Two separate clips per fuse / PCB; quantity is clips, not pairs	Quote	Quote
26	20	761-M1257-3005-N M1257-3005-N	M3 x 0.5 nylon female/female standoff, 10 mm Four insulating feet per PCB	Quote	Quote
27	20	534-29341 29341	M3 x 6 mm nylon screw Four screws per PCB; attach standoff feet from above	Quote	Quote

C. Power, cables and shared supplies

#	Qty	Mouser SKU / Mfr. part	Description / allocation	Unit \$	Ext. \$
28	5	709-GSM18U05-P1J GSM18U05-P1J	Regulated 5 V / 3 A wall supply — US plug One regulated 5 V / 3 A US-plug supply per unit	Quote	Quote
29	5	485-368 368	5.5 × 2.1 mm female barrel-to-screw-terminal adapter One female barrel adapter per unit; short wires connect to PCB J1	Quote	Quote
30	10	490-SDR-70 SDR-70	7-pin DIN male cable plug — solder terminals Two full-size 7-pin male plugs per unit; custom cables	Quote	Quote
31	10	545-P222006 P222-006	PS/2 male-to-female extension cable — 6 ft One male PS/2 end per cable; two cables per unit	Quote	Quote
32	1	485-1311 1311	22 AWG solid-core hookup wire — six-color spool set ONE six-color 22 AWG wire set shared across five builds	Quote	Quote
33	1	474-PRT-09353 PRT-09353	Heat-shrink assortment ONE heat-shrink assortment shared across cable joints	Quote	Quote
34	1	485-4474 4474	USB-A to USB-C data cable — approximately 1 m ONE programming cable shared across five modules	Quote	Quote
35	1	485-850 850	Multimeter for DC voltage, resistance and continuity ONE shared bench multimeter; remove only if already owned	Quote	Quote

Order subtotal / tax / total: pending supplier quotation.

Assembly and purchasing notes

Outside this Mouser PO

Supply 5 hiBCTR RP2040-Zero modules in total (reuse any already owned) and 5 fabricated Rev D PCBs. Module: Amazon ASIN B0DXL12W59. PCB fabrication files are in hardware/rev-d/. No enclosure has been specified; this order builds open boards on insulating feet.

Module sockets: specific orientation

Each PCB uses two SSW-109-01-T-S side sockets and FIVE SSW-101-01-T-S single-contact sockets at M1 contacts 10-14. Turn each single socket so its 3.05 mm housing dimension runs parallel to the long side sockets and its 2.41 mm dimension runs along the five-contact end row. Use the module to align them before soldering. A complete SSW-105 strip has end overhang that nominally overlaps the side sockets at the corners; it is deliberately not on this PO. Sample-fit the actual module and sockets before assembly.

Header pack allocation

One Adafruit 392 pack contains ten 36-pin strips. Cut one strip per board into 9 + 9 + 5 male module contacts and a 2-pin JP1 header. Five strips cover all five units (125 contacts used); five strips remain spare. Do not order five packs.

Fuse clips

Order 10 individual 01110501Z clips for five fuse positions, plus the five 1.5 A fast 5x20 mm fuses. Fit each pair with retaining ears toward the outer ends of the fuse. The clips match the footprint family identified in the saved PCB. No inline holders are needed.

Power and custom cables

Use one regulated 5 V supply and one barrel adapter per unit. Connect adapter + to PCB J1.1 and - to J1.2; the PCB fuse is already in series. Use ten P222-006 cables because each has only one male PS/2 end. Cut about 3 ft from each male end and solder one SDR-70 plug using the Rev D cable table. Continuity-test contacts; do not assume wire colors. Use the plug clamps for strain relief.

Shared supplies and tools

Wire, heat-shrink, header pack, programming cable and multimeter are shared across all five builds. Soldering iron, electronics solder, cutters/stripper, screwdriver and heat source are existing workshop supplies, not included purchase lines. Firmware testing also needs a computer and suitable test equipment. Existing keyboard/mouse receivers and X16 are not part of this order.

Test access and prototype status

TP1-TP26 are bare plated PCB holes, so no separate test-point parts are required. This purchase list does not establish physical fit or a working product: Rev D remains an engineering prototype, with firmware and bench validation pending.

Sources: [Samtec SSW dimensions](#); [Mouser USA contact](#). All SKU links in the tables are clickable.